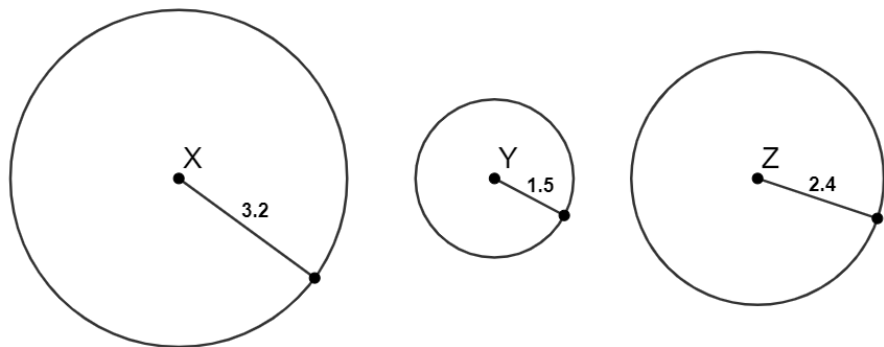


Circles X , Y , and Z are shown. The radius of Circle X is 3.2 units long. The radius of Circle Y is 1.5 units long. The radius of Circle Z is 2.4 units long.



1. Complete the following table. Round to the nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference

2. Complete the table.

Circle X and Y	
Ratio of Radius of Circle X to Radius of Circle Y	
Ratio of Circumference of Circle X to Circumference of Circle Y	

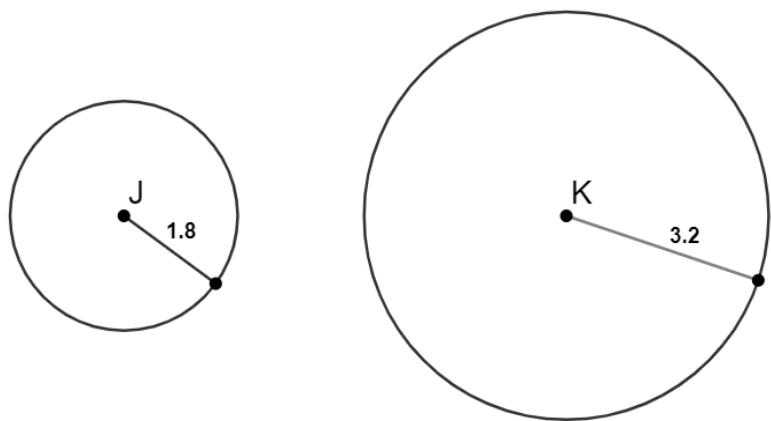
3. Complete the table.

Circle Y and Z	
Ratio of Radius of Circle Y and Radius of Circle Z	
Ratio of Circumference of Circle Y and Circumference of Circle Z	

4. Complete the table.

Circle X and Z	
Ratio of Radius of Circle X to Radius of Circle Z	
Ratio of Circumference of Circle X to Circumference of Circle Z	

Circles *J* and *K* are shown. The radius of Circle *J* is 1.8 units long. The radius of Circle *K* is 3.2 units long.



Show Circle *J* is similar to Circle *K* by completing the proof by contradiction:

5. Assume that Circle *J* is _____ to Circle *K*.
6. The ratio of the radius of Circle *J* to the radius of Circle *K* is _____.
7. The ratio of the circumference of Circle *J* to the circumference of Circle *K* is _____.
8. The ratio of the radii and the ratio of the circumferences are _____, so corresponding parts of Circle *J* and Circle *K* are proportional.
9. Since the corresponding parts of Circle *J* and Circle *K* are proportional, the circles are _____. This is a contradiction, so the assumption that Circle *J* is not similar to Circle *K* is not true.
10.

All
Some
No

 circles are similar.